Fairness Principles as Routing Constraints: How Five Rules for Distributing Urban Air Mobility Noise Respond to Fleet Size

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***Abstract*—The routing objective of an urban air mobility (UAM) network decides which communities bear its noise. We encode five principles of distributive justice (Equal, Sufficientarian, Prioritarian, Utilitarian and Proportional) as hard constraints and lexicographic objectives in one optimising SMT routing solver and measure how each responds to fleet size. On a 5×5 map of Austin, TX with three vertiports and a 6 dB cap on added noise, fleets run from $N = 6$ to 24 aircraft, each cell solved under four aircraft orderings to measure the solver's indifference; Utilitarian and Proportional run a count abstraction of their decibel objective. The principles differ in where noise goes: Proportional confines the fleet to 6 of 25 tiles and cuts summed added noise from 37.7 to 19.7 dB at $N = 16$; Prioritarian holds three protected communities at 0 dB on 8–9 tiles. What they share is the cap's: from $N = 14$ every principle holds the most-exposed community at exactly 4.77 dB above ambient, and above $N = 16$ each pays 3 aircraft-timesteps more than the baseline: the cap conditions and Prioritarian in distance, Utilitarian and Proportional in departure delay. The five produce three distinct footprints: Equal, Sufficientarian and Utilitarian's count form stay within the baseline's; only Prioritarian's and Proportional's objectives narrow it. Of the decibel runs that started, the exact encoding proves Utilitarian to $N = 16$ and Proportional to $N = 8$; the count abstraction proves every cell in every ordering within 472 s, at 16–17 % for Proportional and 86–118 % for Utilitarian on the decibel objective.**

***Index Terms*—urban air mobility, fairness, distributive justice, satisfiability modulo theories, multi-agent routing, environmental noise**

I. Introduction

Urban air mobility (UAM) would move passengers across a city in small electric aircraft at low altitude [1], and community noise is among the factors most likely to decide whether such a network may operate [2]. Aircraft fly between fixed vertiports, so the communities beneath the corridors absorb the noise, and the routing objective decides which communities those are. The background is already unequal: noise exposure in the United States, civil aircraft noise included, is higher in lower-income and minority neighbourhoods [3], [4].

Noise-aware routing formulations usually settle the distributional question implicitly, by minimising total exposure or by holding every receptor below a threshold. Political philosophy gives several distinct accounts of a fair division of a burden [5]–[8], and each can be written as a constraint or an objective over the same routing variables. Gao et al. [9] trade efficiency against a parameterised equity criterion in UAM traffic management; Hauptvogel et al. [10] review distributive rules for aircraft noise without committing them to a planner. We fix five named principles, encode each in one optimising SMT solver [11], [12] and compare their routings on one instance. The encoding follows compilation-based multi-agent path finding [13], [14], with one change: the aircraft do not conflict with each other but are coupled through a shared, nonlinear noise externality.

A principle's effect depends on traffic. A cap on added noise constrains nothing while the fleet fits under it, and a detour costs nothing while the direct corridors have room; both change as the fleet grows. Whether a principle changes the routing, what it costs and where it puts the noise are therefore functions of the fleet size N, and a comparison at one N reports one point on each. This paper measures those functions on a 5×5 map of Austin, TX, for five principles and a no-fairness baseline, from $N = 6$ to $N = 24$.

The scale follows from what is measured. Every number here is a proven optimum, tested against the band four aircraft orderings leave, which needs an instance small enough to close each cell and large enough for the cap to bind; a heuristic planner reaches hundreds of agents but returns one feasible routing, from which neither can be read. What carries to a larger map is the rule, not the size: a cap acts once demand on a tile passes the capacity (3) grants it (Section IV-D).

Its contributions are four measurements:

1) routing and congestion against N, and the three distinct footprints the five principles produce (Section IV-B);
2) travel time and added noise against N, each difference between conditions tested against the solver's indifference among equally optimal routings, and explained through the closed-form capacity a decibel cap grants a tile (Sections IV-C and IV-D);
3) the solve time of each principle across the sweep, on which every cell is proven optimal (Section IV-E);
4) the trade-off between the exact decibel encoding of Utilitarian and Proportional and a count abstraction of it: the loss on the decibel objective against solve time and reach, and the routing each encoding returns (Section IV-F).

II. Five Principles

The burden is the noise aircraft add to a community's ambient level, and a principle chooses among routings rather than among arbitrary allocations. We state each principle's general claim and its reading for routing; Table I gives what the solver receives, in the notation of Section III.

TABLE I

CONDITIONS AS THE SOLVER RECEIVES THEM: THE CAPS SENT AND THE TIER-1 OBJECTIVE, MINIMISED BEFORE THE ARRIVAL PENALTY (5). FOR UTILITARIAN AND PROPORTIONAL THE DECIBEL FORM IS THE EXACT ENCODING AND THE COUNT FORM ITS ABSTRACTION, WHICH PRICES $c_{m,t}$ IN PLACE OF $\delta_m(c_{m,t})$. $\ell = 6$ dB; A CAP NOT SENT IS 30 dB, WHICH NO TILE ON THIS MAP CAN REACH.

Condition	Caps Sent	Tier-1 Objective
Equal	$U{=}\ell$, $D{=}\ell$	none
Sufficientarian	$D{=}\ell$	none
Prioritarian	$D{=}\ell$	$\sum_t \sum_{p\in P} \delta_p(c_{p,t})$
Utilitarian (dB)	$D{=}\ell$	$\sum_t \sum_m \delta_m(c_{m,t})$
Utilitarian (count)	$D{=}\ell$	$\sum_t \sum_m c_{m,t}$
Proportional (dB)	$D{=}\ell$	$\sum_t \sum_m \pi_m\, \delta_m(c_{m,t})$
Proportional (count)	$D{=}\ell$	$\sum_t \sum_m \pi_m\, c_{m,t}$
Baseline	none	none

Equal: everyone receives the same share [15]; added noise should be as equal as possible across communities. *Sufficientarian*: everyone should reach a threshold, above which differences carry no weight [7]; no community's added noise may exceed a limit. *Prioritarian*: a gain counts for more when it goes to the worst-off [5], [6]; the communities already bearing the most burden should receive the least added noise. *Utilitarian*: total harm should be minimal [16]; the fleet's aggregate burden should be as low as possible. *Proportional*: burden should follow benefit [17], [18]; the communities that gain most from the network should carry more of its noise.

Equal and Sufficientarian are encoded as caps and contribute no objective. Prioritarian needs a protected set P and Proportional a benefit price π_m, both inputs from outside the noise model (Section III-C). Utilitarian and Proportional each have two encodings, and the sweep runs their count form (Section III-D). Equal is comparative and Sufficientarian absolute, yet under one limit ℓ they coincide on this instance. Every fleet in this study has $N < M = 25$, so some tile is always unoccupied, the smallest added noise is zero and the spread equals the maximum. The two principles have the same feasible set and the same empty objective.

III. ENCODING AND INSTANCE

A. Routing model

Airspace is $M = 25$ ground tiles $\mathcal{M}$, one per community, tile m carrying ambient level b_m. A fleet of N aircraft is routed over timesteps $t = 0, \ldots, T-1$; aircraft i has origin s_i, destination e_i and position $L_{i,t} \in \mathcal{M}$, with $L_{i,0} = s_i$ and each step a move to one of the eight neighbouring tiles or a hold. An aircraft is audible from its first move through the timestep it lands, marked by the flag $\mathrm{ac}_{i,t}$; an aircraft waiting at its origin or already landed is silent. Tile m then carries

$$c_{m,t} = \sum_i [\, L_{i,t} = m \wedge \mathrm{ac}_{i,t} \,] \tag{1}$$

audible aircraft at time t.

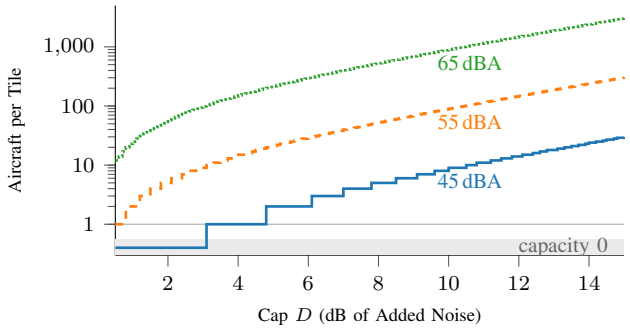


Fig. 1. Aircraft a tile can hold under a cap D on added noise, by ambient level, from (3) with $\alpha = 45$ dBA. At the study's $D = 6$ dB: 2, 29 and 298. Below $D = 3.01$ dB a 45 dBA tile holds none.

B. Noise and the caps

Aircraft noise adds to ambient noise in energy. With per-aircraft source level α, n simultaneous overflights add

$$\delta_m(n) = 10 \log_{10}\!\left(1 + n\, 10^{(\alpha - b_m)/10}\right) \tag{2}$$

decibels to tile m. δ_m is increasing and concave in n and decreasing in b_m: at $\alpha = 45$ dBA one aircraft adds 3.01 dB to a 45 dBA tile and 0.04 dB to a 65 dBA one. The solver tabulates δ_m at centi-decibel resolution.

The absolute cap D requires $\delta_m(c_{m,t}) \le D$ for every m and t. Because δ_m is increasing, it is asserted as the count bound $c_{m,t} \le \kappa_m$ with capacity

$$\kappa_m = \max\{n : \delta_m(n) \le D\} = \left\lfloor \left(10^{D/10} - 1\right) 10^{(b_m - \alpha)/10} \right\rfloor, \tag{3}$$

up to the rounding of the table. The spread cap U requires $\max_m \delta_m(c_{m,t}) - \min_m \delta_m(c_{m,t}) \le U$ at every t, encoded with one upper and one lower auxiliary bound per timestep.

Fig. 1 plots (3). At $\alpha = 45$ dBA and $D = 6$ dB a tile holds 2 aircraft at 45 dBA, 29 at 55 dBA and 298 at 65 dBA; at $D = 10$ dB it holds 9, 90 and 900. One cap on added noise is a hundredfold difference in capacity between the quietest and the loudest tier, and it fills the quiet tiles first.

C. Instance and operating point

The instance is a 5×5 grid over Austin, TX, with 8 tiles at 45 dBA, 12 at 55 dBA and 5 at 65 dBA. Tiles are numbered row-major from the top left, 1 to 25; the ambient map reads, by row, 45 55 55 55 45 / 45 45 45 55 45 / 65 55 55 45 55 / 65 45 65 55 65 / 55 55 55 55 65 dBA, and Fig. 2 shades each tile by it. Vertiports sit at an airport and a downtown core, stations 11 and 18, both 65 dBA, and at an exurban commuter town, station 5, at 45 dBA. Aircraft fly both directions of the three vertiport pairs: at demand factor $f = 1$, 4 per direction between airport and downtown and 2 on each pair involving the town, $N = 16$. Scaling f from 0.25 to 1.5 gives the fleet-size sweep $N \in \{6, 8, 14, 16, 22, 24\}$. The source level is $\alpha = 45$ dBA and a timestep is 180 s.

The protected set P is stations 4, 8 and 14, three low-income communities on the vertiport corridors. The Proportional price is

$$\pi_m = \gamma^{\sum_f \rho_f \, \sigma_f(m)}, \tag{4}$$

with base $\gamma = 5$, factor scores $\sigma_f(m) \in [0, 3]$ for income, ambient level, distance to the nearest vertiport and population, and rank weights $\rho = (0.4, 0.3, 0.2, 0.1)$. On this instance the prices are, by row, 27.09 5.42 13.13 31.83 14.23 / 40.52 10.32 71.17 13.13 15.43 / 8.78 9.52 3.62 90.60 13.13 / 12.12 15.43 1.62 9.52 8.10 / 6.37 43.91 9.52 13.13 18.12, and the solver uses them to two decimals. A low price marks a community that benefits from the network (higher income, louder, nearer a vertiport, less populated), and the objective loads it with traffic.

Every principle faces $D = 6\,\text{dB}$, so the five share one hard constraint and differ in tier 1. The shipped settings, $D = 10\,\text{dB}$ for Equal and Sufficientarian and a protected set of the three loudest non-vertiport tiles (16, 20 and 25), leave the principles little to act on: no cap fills more than 0.44 of a tile at $N = 16$, and the baseline never enters that set at any size. A screen over both, four orderings per cell, selected $D = 6\,\text{dB}$ and the corridor set, under which the cap runs at full capacity from $N = 14$ (Section IV-B).

The *baseline* is the same fleet with both caps at 30 dB and no tier-1 objective: the arrival-optimal routing without a principle.

D. Objectives

Each condition minimises two tiers lexicographically. Tier 1 is the principle's objective (Table I). Tier 2, minimised only among tier-1 optima, is the arrival penalty

$$J_{\text{time}} = \textstyle\sum_i \sum_t [\, \neg \text{ha}_{i,t} \,], \tag{5}$$

the aircraft-timesteps before arrival, where $\text{ha}_{i,t}$ marks aircraft i as landed; time held at the origin counts. Arrival cannot cost a unit of tier 1, and it selects among the routings that tie on it.

The conditions form two lanes. In the *cap lane*, Equal and Sufficientarian have no tier-1 objective, so among cap-feasible routings arrival alone decides. In the *objective lane*, Prioritarian, Utilitarian and Proportional face the same cap and minimise their own objective first. Prioritarian's objective sums added decibels over P only. Utilitarian's and Proportional's price every audible aircraft-timestep, and neither charges an aircraft waiting on the ground.

Utilitarian and Proportional each have two encodings. The *decibel form* prices $\delta_m(c_{m,t})$, the quantity the principle is about, and is the exact encoding. The *count form* prices $c_{m,t}$ instead: $\sum_{t,m} c_{m,t}$ is the fleet's airborne time and $\sum_{t,m} \pi_m c_{m,t}$ its price-weighted counterpart. The abstraction drops the shape of (2). It no longer separates a quiet tile from a loud one, where one aircraft adds 3.01 dB and 0.04 dB respectively, and it charges a tile-timestep the same however many aircraft share it, losing the discount concavity gives for stacking them. Both forms face the same cap, so the capacity asymmetry of (3) constrains both. Section IV-F measures what the abstraction costs and what it buys.

E. Solving

The horizon starts one timestep above the Chebyshev lower bound on trip length (at the bound for Utilitarian). Phase 1 raises it where the caps need more; phase 2 fixes it and optimises both tiers with Z3's νZ optimiser [11], [12], with a landed aircraft required to stay landed, so aircraft can stagger their landings within the horizon.

IV. Results

A. Setup

The five principles and the baseline run at every size of the sweep, from $N = 6$ to 24, on the operating point of Section III-C, with Utilitarian and Proportional in count form (Table I); the nominal demand, $N = 16$, is marked in every plot. No solver timeout was set, and every cell of the sweep is proven optimal. The decibel forms run on the same orderings in Experiment 5, where a run not proven within a harvest horizon of $H = 20.0\,\text{h}$ was stopped and its verified routing kept. Of their 48 runs, 27 started within it, and every decibel statement is over those. Experiments 1–3 report them where an ordering proved: at $N = 6$ and 8 for both, and at $N = 14$ and 16 for Utilitarian.

A solve returns one member of a set of equally optimal routings, and which member depends on the order in which the aircraft are listed. Every cell is therefore solved under four orderings of the same fleet. The feasible set and the optimum do not change, so the range over the four is the solver's indifference at that cell, drawn as a band. Ordering k is the same for every condition. A difference between two conditions is *attributable* when their means differ by more than the larger band and their ranges are disjoint; a smaller difference is stated as within the solver's indifference.

B. Experiment 1: routing and congestion

How do each principle's routing and tile occupancy change as the fleet grows? Fig. 2 draws the routings at every size of the sweep, and the left column of Fig. 3 the peak occupancy and the overlaps.

Congestion grows with N under every condition: peak aircraft over one tile from 1–3 at $N = 6$ to 11–14 at $N = 24$, and overlaps, the tile-timesteps carrying two or more aircraft, from 0–3 to 15–18. On both metrics every principle stays within the solver's indifference of the baseline at every size, with the two exceptions below. The fleet sets how much congestion there is; the principle sets where it sits.

Equal and Sufficientarian pose the same problem at every size of the sweep (Section II), so every difference between them is a difference between members of one optimal set; their arrival penalties agree at every size. At $N = 6$ and 8 the cap is not full, at capacity utilisation 0.50–0.75. From $N = 14$ it binds on the 45 dBA tiles, utilisation 1.00 in every ordering, and from $N = 22$ the traffic spreads onto neighbouring tiles (Fig. 2, $N = 24$): a 45 dBA tile admits two aircraft at a timestep, so a third routes around it, at the 2.75–3.00 extra fleet-hops of Section IV-C. Sufficientarian carries



Fig. 2. Routings under each condition (rows) at every size of the sweep (columns), with Utilitarian and Proportional in both their decibel (dB) and count forms. Tiles are shaded by ambient level, vertiports ringed, and Prioritarian's protected set outlined in its row. Each map draws one of the four aircraft orderings, a proven one where any proved. Line width is proportional to the aircraft on each leg summed over the horizon, so two maps can look alike while their routings differ in timing. Titles give peak added noise (dB) and extra fleet-hops over the shortest-path floor, averaged over the proven orderings. *Timeout*: no ordering of that decibel cell proved within $H = 20.0$ h; the faint map is the verified routing its search held when stopped.

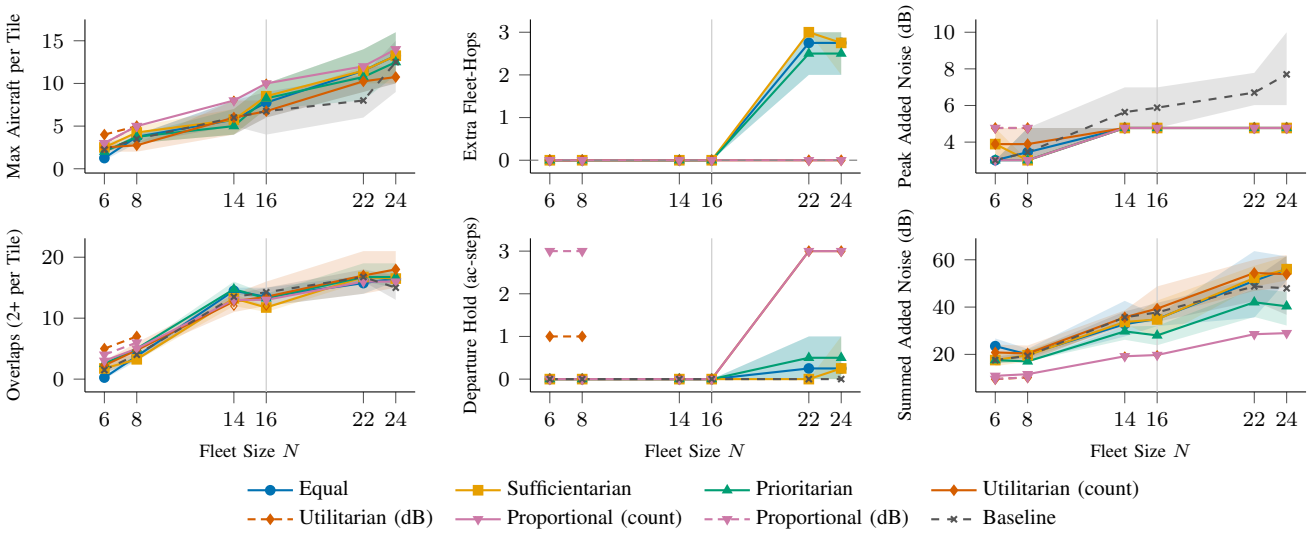


Fig. 3. Congestion (left), travel time (middle) and added noise (right) against fleet size. Lines join the mean over four aircraft orderings; the band is their range, the solver's indifference. Densely dashed lines are a principle's decibel form, joined where all four orderings proved and hollow where only some did (k/4 in Table III); its holds of 12 and 16 at $N = 14$ and 16 are off that panel's scale (Section IV-C). Departure hold counts the aircraft-timesteps a fleet waits beyond its own size, so departing at once scores zero. The vertical line marks the nominal demand, $N = 16$.

fewer overlaps than the baseline at $N = 16$ (11.75 against 14.25).

Prioritarian holds the protected set at 0.00 dB of added noise at every size in every ordering. Staying off three corridor tiles narrows its footprint to 8–9 tiles against the baseline's 10–12, attributably at $N = 6$, 14 and 16, and from $N = 22$ it detours around the protected tiles.

Utilitarian's count objective, airborne time, is minimised by flying every aircraft on a shortest path without hovering, which the baseline does at every size. Under the cap Utilitarian attains the same floor at every size, so its tier 1 keeps exactly the cap-feasible routings on shortest paths and arrival chooses among them. Its congestion stays within the baseline's band at every size.

Proportional confines the fleet to the same 6 tiles at every size, the three vertiports and the three 55 dBA tiles between them (stations 9, 12 and 13), against the baseline's 10–12, attributably from $N = 6$ to 22. It loads them: peak occupancy is 3, 5, 8, 10, 12 and 14 aircraft from $N = 6$ to 24, identical in all four orderings, and attributably above the baseline at $N = 8$ (5 against 3.5). Where they are proven, at $N = 6$ and 8 for both and at $N = 14$ and 16 for Utilitarian, the decibel forms of both objective principles also confine the fleet to 6 tiles, and occupy fewer tile-timesteps than their count forms.

C. Experiment 2: travel time

What does each principle cost against the baseline, in distance and in departure delay? The middle column of Fig. 3 gives both; the arrival penalty (5) counts both.

As run here, every principle is free in travel time through $N = 16$. At $N = 6$, 8, 14 and 16 each attains the baseline's arrival penalty of 18, 22, 40 and 44 aircraft-timesteps in every ordering, with every aircraft on a shortest path. At $N = 22$ and 24 every principle pays exactly 3 aircraft-timesteps more than the baseline (65 against 62, 69 against 66), with zero band.

Equal, Sufficientarian and Prioritarian pay mainly in distance: 2.75, 3.00 and 2.50 extra fleet-hops at $N = 22$ and 2.75, 2.75 and 2.50 at $N = 24$, each attributable. Utilitarian and Proportional fly no extra hop at any size and pay in departure delay: 25 and 27 aircraft-timesteps before first moves at $N = 22$ and 24, against the baseline's 22 and 24, with zero band. The objectives account for the split. Arrival cannot tell a detour from a hold on the ground, and Prioritarian's objective prices only the three protected tiles; Utilitarian's and Proportional's price every airborne aircraft-timestep and none on the ground, so for them a detour costs tier 1 and a hold does not. Their decibel forms pay at every size where they prove: at $N = 6$ and 8 they hold 1 (Utilitarian) and 3 (Proportional) aircraft-timesteps more on the ground than the baseline, and Utilitarian's holds 12 and 16 more at $N = 14$ and 16 in the orderings that proved, all with no extra hop.

D. Experiment 3: added noise and capacity

Where does each principle put the noise as the fleet grows? The right column of Fig. 3 gives the peak and the summed added noise, and Table II the capacity used per ambient tier.

The capacity asymmetry of (3) decides where the cap acts. At $D = 6$ dB a 45 dBA tile holds 2 aircraft and a 65 dBA tile 298, so the cap binds on the quiet tier alone: at $N = 24$ the 45 dBA tiles run at 0.88–1.00 of capacity under every principle as run here and the 65 dBA tiles at 0.02. The cap begins to act where peak demand on a quiet corridor tile passes $\kappa_{45} = 2$: the binding size follows from κ_{45} and the fleet share such a tile carries, and is $N = 14$ here. The shipped settings test that: with $D = 10$ dB and $\kappa_{45} = 9$ over the same sizes, the

TABLE II

CAPACITY UTILISATION BY AMBIENT TIER: PEAK AIRCRAFT OVER A COMMUNITY AS A FRACTION OF THE CAPACITY (3) GRANTS A TILE AT THAT AMBIENT LEVEL, AVERAGED OVER THE TIER'S TILES THAT CARRIED TRAFFIC, FOR THE FIRST AIRCRAFT ORDERING. *Max* IS THE LARGEST UTILISATION ON ANY TILE, AVERAGED OVER THE ORDERINGS. EVERY ROW SENDS D=6, WHICH GRANTS 2/29/298 AIRCRAFT AT 45/55/65 DBA. A DAGGER MARKS A DECIBEL-FORM CELL NOT PROVEN IN ALL FOUR ORDERINGS WITHIN $H = 20.0$ H; ITS VALUES COME FROM THE ORDERINGS THAT RAN, PROVEN OR STOPPED AT H (SEC. IV-F).

N	Condition	45 dBA	55 dBA	65 dBA	Max
8	Equal	0.50	0.06	0.01	0.62
8	Sufficientarian	0.50	0.06	0.01	0.50
8	Prioritarian	0.50	0.08	0.01	0.50
8	Utilitarian (dB)	1.00	0.13	0.01	1.00
8	Utilitarian (count)	0.50	0.06	0.01	0.75
8	Proportional (dB)	1.00	0.13	0.01	1.00
8	Proportional (count)	0.50	0.11	0.01	0.50
24	Equal	0.92	0.19	0.02	1.00
24	Sufficientarian	0.92	0.19	0.02	1.00
24	Prioritarian	0.88	0.22	0.02	1.00
24	Utilitarian (dB)†	0.67	0.13	0.02	1.00
24	Utilitarian (count)	0.92	0.13	0.02	1.00
24	Proportional (dB)†	1.00	0.32	0.03	1.00
24	Proportional (count)	1.00	0.32	0.02	1.00

quiet tier runs at 0.11–0.39 of capacity and no tile reaches it before $N = 24$, where one Equal routing fills one at 10.00 dB. The decibel forms fill the quiet tier at every size where they prove, from $N = 6$: 1.00, against 0.50–0.80 for their count forms (Table II). Concavity is the reason: two aircraft over one 45 dBA tile add 4.77 dB, one over each of two add 6.02, so the decibel objective stacks them up to the cap.

Peak. At $N = 6$ and 8 the principles' peaks lie between 3.01 and 3.89 dB, within the baseline's band; one aircraft over a quiet tile adds 3.01 dB. From $N = 14$ every principle returns exactly 4.77 dB in every ordering at every size: two aircraft over a 45 dBA tile, the most (3) admits there. Over $N = 14$, 16, 22 and 24 the baseline's peak averages 5.64, 5.88, 6.70 and 7.70 dB with bands of 2.22, 2.22, 1.76 and 3.98 dB. The separation clears the band at $N = 22$ for all five principles.

Summed noise. Proportional's summed added noise has zero band at every size: 10.9, 11.6, 19.2, 19.7, 28.5 and 29.0 dB from $N = 6$ to 24, against the baseline's 17.6, 19.4, 35.7, 37.7, 48.8 and 47.9 dB. The difference is attributable at every size but $N = 24$, where the baseline's band is 24.9 dB wide. Proportional occupies 13 tile-timesteps at $N \leq 16$ against the baseline's 16–19 with the same airborne time, and since δ_m is concave, fewer occupied tile-timesteps add less in sum. Prioritarian's summed noise is attributably lower at $N = 14$ and 16 (29.7 and 28.0 against 35.7 and 37.7 dB). Equal, Sufficientarian and Utilitarian's count form stay within the baseline's band at every size. The decibel forms add 9.54 (Utilitarian) and 9.60 dB (Proportional) summed at $N = 6$, against the baseline's 17.6, and already peak at 4.77 dB at $N = 6$ and 8. At $N = 14$ and 16 Utilitarian's decibel form sums to 19.2 and 19.7 dB in the orderings that proved, against 35.7 and 39.3 in count form. Proportional's count form sums to the same within 0.01 dB and arrives 12 and 16 aircraft-timesteps earlier: tier 1 takes any saving in noise over any delay.

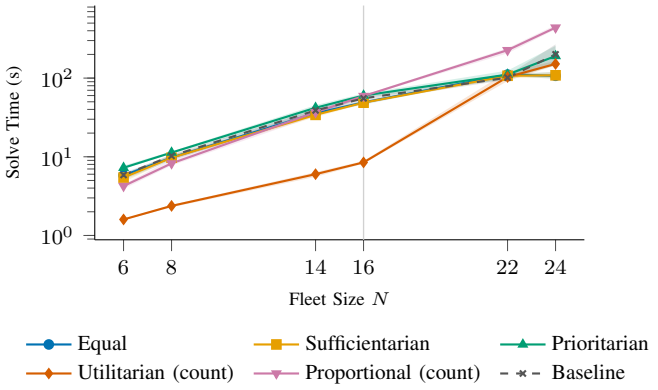


Fig. 4. Mean wall time to a proven optimum over four orderings, eight solves in parallel; the band is the range over the orderings. Utilitarian and Proportional are in count form; Fig. 5 gives their decibel forms.

E. Experiment 4: solver performance

How does solve time grow with the fleet? Fig. 4 gives the mean wall time to a proven optimum over the four orderings, with eight solves running in parallel on one 14-core workstation. Every condition here proves every cell. Equal, Sufficientarian and Prioritarian solve within a factor of two of the baseline at every size, and the baseline takes 5.9 s at $N = 6$ and 200 s at $N = 24$. Utilitarian in count form is the fastest condition through $N = 16$, at 1.6 s for $N = 6$ and 8.5 s for $N = 16$ against the baseline's 5.9 and 55.3 s. We attribute that to its tier 1, which excludes every detour before arrival is searched, while the baseline must weigh them because a detour and a hold cost it the same. Proportional in count form is the slowest at $N = 22$ and 24, at 227 and 439 s. Prioritarian's tier 1 is a decibel sum too, but over its three protected tiles, and it reaches zero at every size; the decibel forms of Utilitarian and Proportional sum over all 25 tiles, and Fig. 5 gives their solve times.

F. Experiment 5: exactness against scalability

What does the count abstraction cost in the principle's own objective and in the noise metrics, and what does it buy in solve time and reach across the sweep? Table III scores both forms' routings on the decibel objective J_{dB}, paired by aircraft ordering. The count-form routing has the same caps and horizon as the decibel problem, so it is feasible for it and can only score at or above the optimum J^*.

Utilitarian. At $N = 6$ and 8 the decibel form proves in 49 and 296 s against the count form's 1.6 and 2.4 s, 30 and 124 times slower. The count-form routings score 20.79 and 20.37 dB on the decibel objective against optima of 9.52 and 10.24 dB: a loss of 11.27 and 10.13 dB, or 118 and 99 %, attributable at both sizes. The count objective, airborne time, is minimised by every shortest-path routing, so the count form leaves the decibel objective to the arrival tie-break.

The decibel form concentrates the fleet: it occupies 10 tile-timesteps against 15.25 and 16.25, and holds one aircraft-timestep more on the ground (7 against 6 and 9 against 8), both attributable. At $N = 14$ and 16, where two and one decibel orderings proved in a mean of 3.5 and 12.3 h, the count routings score 35.66 and 39.31 dB against optima of 19.18 and 19.72: 86 and 99 %, attributable.

Proportional. At $N = 6$ and 8 the decibel form proves in a mean of 2655 and 17 793 s against the count form's 4.2 and 8.2 s, 625 and 2159 times slower. The count-form routing scores 128.00 and 134.43 price-weighted dB in every ordering against optima of 109.43 and 115.85: a loss of 18.58 at both sizes, or 17 and 16 %, attributable. The decibel form lowers the sum and raises the peak. It adds 9.60 and 10.31 dB summed against 10.92 and 11.63 and occupies 11 tile-timesteps against 13, while its peak is 4.77 dB against 3.01 and it holds three aircraft-timesteps more on the ground (9 against 6, 11 against 8), all attributable.

Reach. Of the 27 decibel runs that started within H, none proved above $N = 16$ for Utilitarian or $N = 8$ for Proportional: 19 proved and 8 were stopped at the harvest with a verified routing. The other 21 runs never started, so every size above $N = 8$ rests on one or two orderings and the frontier is a statement about the runs that ran, not a bound on the encoding. For seven of the eight stopped runs Z3 reported no finite lower bound on the decibel objective, so no gap to the optimum can be quoted; the eighth, Utilitarian at $N = 16$, had closed its gap at the optimum another ordering proved. Against the best decibel routing any run reached, which lies at or above the optimum, the count routings lose at least 55 and 40 % for Utilitarian at $N = 22$ and 24, and at least 1.3 and 1.2 % for Proportional at $N = 14$ and 16. At $N = 22$ and 24 no decibel run reached a routing better than Proportional's count form: at $N = 24$ the count routing scores 331.37, below the 334.30 the decibel form reached in 14.8 h. The count forms prove every cell to $N = 24$, the slowest in 472 s.

The trade-off. The exact encoding proves only at the smallest sizes, in 49 s to 12.3 h; the abstraction proves every size in minutes, 30 to 2159 times faster where both prove. Proportional's count form keeps the benefit price and loses 16–17 % where the optimum is known, staying within 1.4 % of the best decibel routing reached above $N = 8$; Utilitarian's reduces to airborne time, which carries no information about noise, and loses 86–118 %.

TABLE III

EXPERIMENT 5. J_{dB} IS THE DECIBEL OBJECTIVE: SUMMED ADDED NOISE IN DB FOR UTILITARIAN, PRICE-WEIGHTED DB FOR PROPORTIONAL. J^* IS ITS OPTIMUM, PROVEN BY THE DECIBEL FORM; *Count* IS ITS VALUE ON THE COUNT-FORM ROUTINGS (MEAN OVER FOUR ORDERINGS), *Loss* THE DIFFERENCE. TIMES ARE MEAN WALL SECONDS TO A PROVEN OPTIMUM. A DAGGER MARKS A DECIBEL CELL NOT PROVEN BY THE HARVEST: J^* IS THEN THE BEST VALUE REACHED (AN UPPER BOUND ON THE OPTIMUM), THE LOSS A LOWER BOUND, AND THE TIME AND SPEED-UP LOWER BOUNDS. A CELL WHERE NO DECIBEL RUN REACHED A ROUTING BETTER THAN THE COUNT FORM'S CARRIES NO POSITIVE BOUND AND IS LEFT OUT: PROPORTIONAL AT $N = 22$ AND 24. A TIME MARKED (k/4) IS THE MEAN OVER THE k ORDERINGS THAT PROVED.

N	J^*	Count	Loss	t_{dB}	t_{count}	Speed-up
Utilitarian						
6	9.52	20.79	11.27 (118%)	49	1.6	30×
8	10.24	20.37	10.13 (99%)	296	2.4	124×
14	19.18	35.66	16.48 (86%)	12560 (2/4)	6.0	–
16	19.72	39.31	19.59 (99%)	44439 (1/4)	8.5	–
22	35.03†	54.38	≥19.35 (≥55.2%)	>60229	103.3	>583×
24	38.34†	53.85	≥15.51 (≥40.5%)	>54322	151.4	>359×
Proportional						
6	109.4	128.0	18.6 (17%)	2655	4.2	625×
8	115.8	134.4	18.6 (16%)	17793	8.2	2159×
14	217.2†	220.0	≥2.8 (≥1.3%)	>51557	36.8	>1401×
16	222.0†	224.8	≥2.8 (≥1.2%)	>68864	59.2	>1164×

TABLE IV

RESPONSE OF EACH PRINCIPLE TO FLEET SIZE: THE CHANGE AGAINST THE BASELINE AND THE SIZES N AT WHICH IT HOLDS. "AS BASELINE" IS WITHIN THE SOLVER'S INDIFFERENCE AT EVERY SIZE, EXCEPT SUFFICIENTARIAN'S FEWER OVERLAPS AT $N = 16$. TRAVEL TIME IS FREE TO $N = 16$ FOR THE CONDITIONS AS RUN AND COSTS 3 AIRCRAFT-TIMESTEPS AT $N = 22$ AND 24; THE COLUMN GIVES THE CURRENCY. THE PEAK IS FIXED AT 4.77 DB WITH ZERO BAND FROM $N = 14$. PRIORITARIAN'S NARROWER FOOTPRINT IS ATTRIBUTABLE AT $N = 6$, 14 AND 16. UTILITARIAN AND PROPORTIONAL ROWS ARE THE COUNT FORMS; THE INDENTED *dB form* ROWS ARE THEIR DECIBEL FORMS, OVER THE SIZES WHERE ALL FOUR DECIBEL ORDERINGS ARE PROVEN. *Wall*: MEAN SECONDS TO A PROVEN OPTIMUM AT $N = 24$; FOR A DECIBEL FORM, THE LARGEST N PROVEN IN ANY ORDERING WITHIN H.

	Congestion	Travel Time	Added Noise	Wall (s)
Equal	as baseline	distance, 22+	peak fixed, 14+	107
Sufficientarian	as baseline	distance, 22+	peak fixed, 14+	109
Prioritarian	8–9 tiles	distance, 22+	P at 0 dB; peak 14+	191
Utilitarian	as baseline	delay, 22+	peak fixed, 14+	151
dB form	6 tiles, 6–8	delay, 6–8	sum 6–8; peak 6–8	$N \leq 16$
Proportional	6 tiles, 6–22	delay, 22+	sum 6–22; peak 14+	439
dB form	6 tiles, 6–8	delay, 6–8	sum 6–8; peak 6–8	$N \leq 8$
Baseline	–	–	set by ordering	200

G. Summary

Table IV collects each principle's response to fleet size.

V. CONCLUSION

We encoded five distributive principles as caps and lexicographic objectives in one routing solver and measured their response to fleet size on one instance, from $N = 6$ to $N = 24$, every cell of the sweep proven optimal under four aircraft orderings.

Equal and Sufficientarian pose one problem across the sweep: they keep congestion within the solver's indifference of the baseline at all but one cell, fix the peak at 4.77 dB once the cap binds at $N = 14$, and pay 2.75 to 3.00 extra fleet-hops for it from $N = 22$. Prioritarian holds its protected communities at 0 dB at every size, narrows the fleet's footprint, lowers summed noise at $N = 14$ and 16, and pays in distance from $N = 22$. Utilitarian, in count form, keeps every aircraft on a shortest path at every size and pays the same price as departure delay; on this instance it differs from Sufficientarian only in preferring a hold to a detour, and only above $N = 16$. Proportional, also in count form, confines the fleet to six tiles with zero band at every size, holds summed added noise

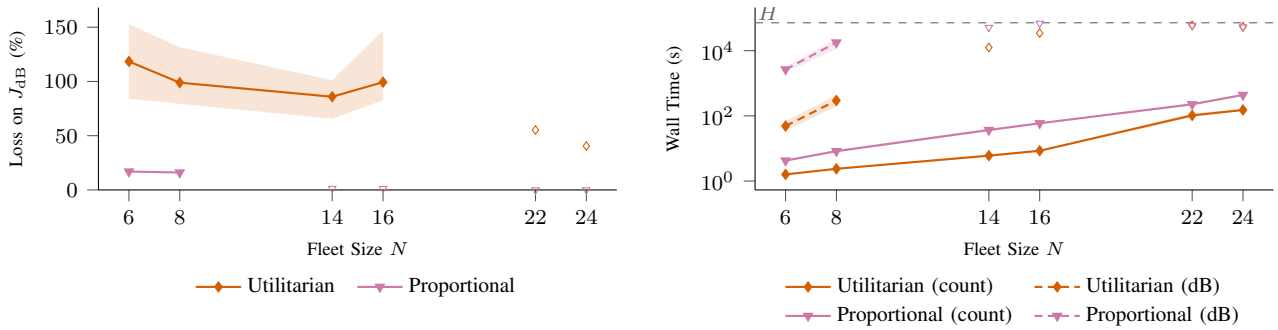


Fig. 5. Experiment 5. Left: loss of the count-form routings on the decibel objective, $(J_{\mathrm{dB}}(\mathrm{count}) - J^*)/J^*$, mean over four orderings with its range; hollow marks a lower bound where no decibel ordering proved. Right: mean wall time to a proven optimum for both forms; hollow is the mean elapsed time of runs not all proven by the harvest, a lower bound; the dashed line is the harvest horizon H.

below the baseline's from $N = 6$ to 22, and also pays on the ground. The baseline's peak varies with the aircraft ordering alone, by up to 3.98 dB. Five principles thus yield three distinct footprints on this instance: Equal, Sufficientarian and Utilitarian's count form never leave the baseline's, and only Prioritarian's and Proportional's objectives narrow it.

For a designer, fleet size decides what the choice of principle buys. Below the size at which the cap fills, only Prioritarian's and Proportional's objectives move the routing, onto fewer tiles. Above it, every principle fixes the peak and pays the same arrival price, and the choice is which communities carry the remaining noise and whether the delay is taken in the air or on the ground.

Utilitarian and Proportional trade exactness for scale. Their decibel encoding prices the noise each principle is about and, of the runs launched, proves Utilitarian to $N = 16$ and Proportional to $N = 8$ within 20 h; the count abstraction proves every cell, the slowest run in 472 s. The abstraction costs 16–17 % on Proportional's decibel objective, which keeps its benefit price, and 86–118 % on Utilitarian's, whose count objective reduces to airborne time and carries no information about noise. The encoding also decides what Utilitarian does to the routing. In decibel form, wherever it proves, it confines the fleet to 6 tiles, fills the quiet tiles from $N = 6$ and pays in departure delay, 16 aircraft-timesteps at $N = 16$, for summed noise that Proportional's count form matches within 0.01 dB at no delay.

The acoustic model is coarse, with uniform tiles and point sources without altitude or directivity [2], and the findings rest on one instance and one cap. Four orderings bound the solver's indifference from below.

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